

The 025 Campaign on the mmli Node

Records, split and gene representation crossed over trigenic interaction prediction: measured cells
filled, planned cells empty

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Abstract

Two claims rest on this campaign. Training on fitness records merged across datasets improves prediction of trigenic interactions that the added records do not contain, and a fixed sequence-derived gene representation, which transfers to genes and organisms a learned table has never seen, costs nothing where the table is at its best. One seed of the first claim is measured: adding 739,219 double-mutant records to the 376,732 triples raises validation Pearson from 0.439 (three seeds, sd 0.003) to 0.478. The second claim has no measurement on the random split. Table 1 holds every cell of the design and Table 2 the order in which the single four-GPU node runs the 26 jobs that fill it, about 23 days end to end.

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

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1 Design ✗

Vocabulary. Every arm is the equivariant cell graph transformer of experiment 010 trained on a subset of the 025 build (13.5 million records) and scored on trigenic interaction alone.

composite A fixed per-gene vector of 3,328 dimensions: promoter (768), open reading frame as codons under CaLM (768), open reading frame as protein under ProtT5 (1,024) and terminator (768), entered through a two-layer preprocessor whose width (339) matches the table’s parameter count to 0.13 percent.

fit. Joint fitness supervision: the model predicts fitness and interaction, both at loss weight 1.0, with a perturbed classification token and a linear readout.

Q The query-pair-disjoint split. A trigenic screen crosses one double-mutant query strain against an array of single mutants; 420 query pairs recur, 89 are held out, and every triple of a held-out pair is in validation or test (37,705 and 37,791).

R The random split of experiment 010, carried onto the 025 build by genotype identity: 376,732 of 376,732 triples matched, 301,386 train, 37,673 validation, 37,673 test.

S0 The 376,732 triples.

S3 S0, the 5,694 singles, and the 739,219 closure doubles, those whose gene pair lies inside some triple: 1,121,645 records. Every record outside the pinned split trains.

S3 strict S3 without the 85 doubles that are themselves a held-out query pair of Q: 1,121,560 records. The 145,208 doubles pairing one held-out query gene with an array gene stay, since digenic screens supply them before any trigenic screen is run.

S4 S0, the singles, and 739,219 random doubles whose pair lies inside no triple. Size-matched to S3 and sharing none of its doubles.

score The mean of validation Pearson on trigenic interaction over epochs 10 to 29, reported only when all 20 epochs are logged. The maximum over epochs is an upward-biased order statistic and is never the score.

table A learned $6,607 \times 180$ gene embedding, 1,189,260 parameters.

Reads declared before the runs. Each planned cell differs from its comparator in one factor, and the Δ column of Table 1 is that difference. S3 against S0 on R with the table tests whether added fitness records help; three seeds all above the S0 range is the criterion, and the S0 seed spread is 0.003. S4 against S3 separates the triples’ own pairs from doubles in general. The stored-value identity linking a triple’s score to its doubles reaches only $r = 0.23$ on this build, so S4 matching S3 is a live outcome and would restate the claim as one about data volume. The composite against the table on R, at S0 and at S3, tests whether the transferable representation matches the table when every gene is seen in training. S3 strict against S0 on Q asks the same data question where query pairs are unseen, under both representations, and whether the table closes on the composite once the held-out triples’ genes appear in training doubles. The single-region arms on Q rank CaLM, ProtT5 and the flanks; one seed each cannot, since seeds of one arm spread by 0.015 to 0.033.

2 The campaign table ✗

Table 1: Every cell of the campaign. A number is a finished seed’s score; *run, ep k* is the running job and the last validation epoch it logged; *qk* is position *k* in the queue of Table 2; n/a marks a seed slot the cell does not use. Seeds a, b, c are 1, 2, 3, except in the rows inherited from the first disjoint round, where they are 42, 1, 2. *n* is seeds scored over seeds planned, Δ the mean minus the comparator cell’s mean (Section 1), and the best mean of each block is bold. Generated by `mmli_campaign_plan.py`.

records	genes	fit.	config	seed a	seed b	seed c	<i>n</i>	mean	sd	Δ
A. Random split (010’s pinned split, 37,673 validation triples)										
S0	table	no	s0_r_kl_ctrl_013	0.438	0.436	0.441	3/3	0.439	0.002	
S0	table	yes	s0_r_kl_fit_014	0.443	0.438	0.437	3/3	0.439	0.003	+0.001
S3	table	yes	s3_r_kl_fit_031	0.478	<i>q1</i>	<i>q2</i>	1/3	0.478		+0.038
S0	composite	yes	s0_r_kl_embfit_035	<i>q3</i>	<i>q4</i>	<i>q5</i>	0/3			
S3	composite	yes	s3_r_kl_embfit_034	<i>q6</i>	<i>q7</i>	<i>q8</i>	0/3			
S4	table	yes	s4_r_kl_fit_039	<i>q14</i>	<i>q24</i>	<i>q25</i>	0/3			
B. Query-pair-disjoint split (89 of 420 query pairs held out, 37,705 validation triples)										
S0	table	no	s0_q_kl_ctrl_016	0.140	0.144	0.125	3/3	0.136	0.010	
S0	random vector	no	s0_q_kl_rand_018	0.151	0.146	0.158	3/3	0.152	0.006	+0.015
S0	composite	no	s0_q_kl_emb_017	0.215	0.235	0.203	3/3	0.218	0.016	+0.081
S0	CaLM alone	no	s0_q_kl_calml_020	0.204	<i>run, ep 17</i>	<i>q11</i>	1/3	0.204		-0.014
S0	ProtT5 alone	no	s0_q_kl_prot_021	0.162	<i>q9</i>	<i>q12</i>	1/3	0.162		-0.055
S0	flanks alone	no	s0_q_kl_fudt_032	0.164	<i>q10</i>	<i>q13</i>	1/3	0.164		-0.053
S0	table	yes	s0_q_kl_fit_038	<i>q15</i>	<i>q16</i>	<i>q17</i>	0/3			
S0	composite	yes	s0_q_kl_embfit_027	0.209	0.237	0.171	3/3	0.206	0.033	-0.012
S3 strict	table	yes	s3_q_kl_fit_036	<i>q19</i>	<i>q21</i>	<i>q23</i>	0/3			
S3 strict	composite	yes	s3_q_kl_embfit_037	<i>q18</i>	<i>q20</i>	<i>q22</i>	0/3			

What is measured. On R, fitness supervision alone adds nothing to the triples: 0.439 with it, 0.439 without. On Q the composite (0.218) separates from the random vector (0.152) and from the table (0.136) with no overlap between their seeds, so sequence content carries to unseen query pairs and a learned table does not.

What is one seed. The S3 seed on R scored 0.478 over epochs 10 to 29 (run yb4gjh51, job 2409033, 104 epochs). Its validation Pearson peaked at 0.498 at epoch 32 and drifted to 0.478 by epoch 100 while train Pearson reached 0.768, which is why every S3 and S4 cell now stops at 50 epochs. Its row carries $n = 1/3$, so the mean and Δ printed there are that seed alone. CaLM alone (0.204), ProtT5 alone (0.162) and the flanks alone (0.164) are one seed each.

3 Queue order and clock ✗

The mmli partition is one node with four A100s and every job takes all four, so the campaign is a single chain in which each job starts when the previous one ends, whether it succeeded or not. An S0 run of 30 epochs takes 7.5 hours (jobs 2408604 to 2408607) and an S3 or S4 run of 50 epochs 33 hours (39.5 minutes per epoch, job 2409033).

The order follows the two claims. The S3 replicates on R come first, then the composite on R at S0 (a day for all three seeds) and at S3, then the single-region replicates on Q, the first S4 seed, the table-with-fitness baseline on Q, the S3 strict cells on Q with the two representations alternating by seed so that a campaign stopped early is still balanced, and the remaining S4 seeds.

Table 2: The chain as the scheduler holds it. Hours are the measured run time of the record subset, less the elapsed time for the running job; the last column accumulates them. Generated by `mmli_campaign_plan.py` from one `squeue` read.

#	job	name	config	state	seed	hours	cumulative h
0	2409034	025-q-calm-s1	s0_q_ql_calm_020	running	1	2.0	2
1	2409672	025-s3-fit2-e50	s3_r_ql_fit_031	pending	2	33.0	35
2	2409673	025-s3-fit3-e50	s3_r_ql_fit_031	pending	3	33.0	68
3	2409705	025-r-emb-s1	s0_r_ql_embfit_035	pending	1	7.5	76
4	2409706	025-r-emb-s2	s0_r_ql_embfit_035	pending	2	7.5	83
5	2409707	025-r-emb-s3	s0_r_ql_embfit_035	pending	3	7.5	91
6	2409708	025-s3r-emb-s1	s3_r_ql_embfit_034	pending	1	33.0	124
7	2409709	025-s3r-emb-s2	s3_r_ql_embfit_034	pending	2	33.0	157
8	2409710	025-s3r-emb-s3	s3_r_ql_embfit_034	pending	3	33.0	190
9	2409035	025-q-prot-s1	s0_q_ql_prot_021	pending	1	7.5	197
10	2409036	025-q-fudt-s1	s0_q_ql_fudt_032	pending	1	7.5	205
11	2409037	025-q-calm-s2	s0_q_ql_calm_020	pending	2	7.5	212
12	2409038	025-q-prot-s2	s0_q_ql_prot_021	pending	2	7.5	220
13	2409039	025-q-fudt-s2	s0_q_ql_fudt_032	pending	2	7.5	227
14	2409711	025-s4r-tab-s1	s4_r_ql_fit_039	pending	1	33.0	260
15	2409712	025-q-tabfit-s1	s0_q_ql_fit_038	pending	1	7.5	268
16	2409713	025-q-tabfit-s2	s0_q_ql_fit_038	pending	2	7.5	275
17	2409714	025-q-tabfit-s3	s0_q_ql_fit_038	pending	3	7.5	283
18	2409715	025-s3q-emb-s1	s3_q_ql_embfit_037	pending	1	33.0	316
19	2409716	025-s3q-tab-s1	s3_q_ql_fit_036	pending	1	33.0	349
20	2409717	025-s3q-emb-s2	s3_q_ql_embfit_037	pending	2	33.0	382
21	2409718	025-s3q-tab-s2	s3_q_ql_fit_036	pending	2	33.0	415
22	2409719	025-s3q-emb-s3	s3_q_ql_embfit_037	pending	3	33.0	448
23	2409720	025-s3q-tab-s3	s3_q_ql_fit_036	pending	3	33.0	481
24	2409721	025-s4r-tab-s2	s4_r_ql_fit_039	pending	2	33.0	514
25	2409722	025-s4r-tab-s3	s4_r_ql_fit_039	pending	3	33.0	547